MQGT-SCF Phase II: Scalar-Singlet EFT Extension, GKSL Measurement Container, and Interferometric Test Program

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We introduce the Merged Quantum Gauge and Scalar Consciousness Framework (MQGT-SCF), an extension of the Standard Model and General Relativity by two real scalar singlets $\Phi_{\rm c}(x)$ and $E(x)$ coupled to quantum measurement dynamics. The framework preserves gauge invariance, Lorentz covariance, and renormalizability at the effective field theory level. We formulate a covariant, completely positive trace-preserving (CPTP) evolution via a Tomonaga–Schwinger-compatible GKSL generator, ensuring consistency with relativistic quantum theory and microcausality. Within this structure, $\Phi_{\rm c}$ and E enter as dynamical scalar degrees of freedom; an optional measure-level bias yields modified effective decoherence weighting, with standard quantum mechanics recovered in the $\eta \to 0$ limit. The flagship testable prediction is interferometric visibility suppression $V/V_0 \sim \exp(-\Gamma T \Delta x^2)$, with exclusion floor $\Gamma_{\rm floor} \approx 1.15 \times 10^9\,{\rm s}^{-1}{\rm m}^{-2}$ at $T = 10^{-6}$ s, $\Delta x = 10^{-3}$ m. We state the minimal EFT action, boundedness conditions, decoupling limit, and explicit failure conditions. While the interpretation of $\Phi_{\rm c}$ and E admits broader philosophical readings, the present work restricts itself to their role as dynamical scalar degrees of freedom within a consistent quantum field-theoretic and open-system formulation.

I. INTRODUCTION

This paper formalizes the Phase II research program for MQGT-SCF. The pivot is explicit: we restore standard unitary microdynamics as the mainline and treat any modified-collapse or Born-tilt construction as a companion hypothesis that must earn survival through data. The framework extends GR+SM by scalar-singlet sector(s) with explicit parameter hygiene and a staged falsification program. This document does not claim empirical confirmation; it defines the structure that reality can test.

II. CORE DECISION AND CANONICAL SPINE

(1) **Standard unitary evolution** governs the underlying microdynamics.

(2) **Measurement**, where operationalized, is expressed within one canonical local, covariant CPTP/GKSL framework that reduces back to ordinary quantum mechanics in the appropriate limits.

(3) **Teleology**, if retained, is implemented as a measure tilt over trajectories or outcomes, not as a primitive time-odd bulk source term in the main Lagrangian.

(4) **MQGT-SCF** is the current effective phenomenological and test-facing layer; deeper SO(10)/string/uniqueness structure remains an exploratory UV program.

(5) **Zora** belongs to the execution, simulation, and agent-architecture layer, not to the minimal physics statement.

(6) The program is taken to be ahead at formal/specification closure, but not yet at decisive external empirical confirmation.

III. MINIMAL EFT ACTION AND FIELD EQUATIONS

The minimal extension adds two real scalar singlets on spacetime $(M, g_{\mu\nu})$: $\Phi_{\rm c}(x)$ and $E(x)$, in addition to GR+SM fields. We work as an effective field theory with local operators of canonical dimension ≤ 4. The action is

$$S = \int {\rm d}^4x\,\sqrt{-g}\,\Big(\mathcal{L}_{\rm GR} + \mathcal{L}_{\rm SM} + \mathcal{L}_{\Phi_{\rm c}} + \mathcal{L}_E + \mathcal{L}_{\rm int}\Big), \quad (1)$$

with

$$\mathcal{L}_{\rm GR} = \frac{M_{\rm Pl}^2}{2} R - \Lambda, \quad (2)$$

$$\mathcal{L}_{\Phi_{\rm c}} = -\frac{1}{2}\left(\nabla\Phi_{\rm c}\right)^2 - \frac{1}{2} m_{\Phi_{\rm c}}^2 \Phi_{\rm c}^2 - \frac{\lambda_{\Phi_{\rm c}}}{4}\Phi_{\rm c}^4, \quad (3)$$

$$\mathcal{L}_E = -\frac{1}{2}\left(\nabla E\right)^2 - \frac{1}{2} m_E^2 E^2 - \frac{\lambda_E}{4} E^4. \quad (4)$$

The minimal renormalizable interactions are

$$\mathcal{L}_{\rm int} = -\frac{\kappa}{2}\Phi_{\rm c}^2 E^2 - \gamma\,\Phi_{\rm c}\,E - \alpha_H |H|^2 \Phi_{\rm c} - \beta_H |H|^2 E - g_{\Phi_{\rm c}} \Phi_{\rm c}\,\mathcal{O}_m - g_E E\,\mathcal{O}_m \quad (5)$$

Here H is the SM Higgs doublet and $\mathcal{O}_m$ is a coarse-grained matter/neural order parameter. The bilinear $\gamma\,\Phi_{\rm c} E$ is treated as a standard interaction.

Varying the action gives the scalar equations of motion

$$\Box\Phi_{\rm c} + m_{\Phi_{\rm c}}^2 \Phi_{\rm c} + \lambda_{\Phi_{\rm c}}\Phi_{\rm c}^3 + \kappa\,\Phi_{\rm c}\,E^2 + \gamma\,E = J_{\Phi_{\rm c}}(x), \quad (6)$$

$$\Box E + m_E^2 E + \lambda_E E^3 + \kappa\,E\,\Phi_{\rm c}^2 + \gamma\,\Phi_{\rm c} = J_E(x), \quad (7)$$

with canonical source terms

$$J_{\Phi_c}(x) = g_{\Phi_c}\,\mathcal{O}_m(x) + \alpha_H |H(x)|^2, \qquad J_E(x) = g_E\,\mathcal{O}_m(x) + \beta_H |H(x)|^2. \tag{8}$$

IV. VACUUM STABILITY AND BOUNDEDNESS GATE

The quartic potential energy is

$$V_4(\Phi_c, E) = \frac{\lambda_{\Phi_c}}{4}\Phi_c^4 + \frac{\lambda_E}{4}E^4 + \frac{\kappa}{2}\Phi_c^2 E^2.$$

A standard sufficient condition for boundedness is

$$\lambda_{\Phi_c} > 0, \qquad \lambda_E > 0, \qquad \kappa > -\sqrt{\lambda_{\Phi_c}\lambda_E}. \tag{9}$$

V. MEASUREMENT: LOCAL COVARIANT CPTP/GKSL CONTAINER

Let Σ be a Cauchy surface and $\rho[\Sigma]$ a density functional. The canonical non-unitary dynamics is local GKSL on hypersurfaces:

$$\frac{\delta\rho[\Sigma]}{\delta\Sigma(x)} = -\frac{i}{\hbar}\,[\mathcal{H}(x), \rho[\Sigma]] + \sum_a \left(L_a(x)\rho[\Sigma]L_a^\dagger(x) - \frac{1}{2}\,\{L_a^\dagger(x)L_a(x), \rho[\Sigma]\}\right). \tag{10}$$

The Lindblad operators $L_a(x)$ are taken as local functionals of the scalar sector and matter operators, constrained by CPTP, locality, and bounded heating; explicit constructions are deferred to companion work.

Microcausality: $[L_a(x), L_b(y)] = 0$ whenever x and y are spacelike separated. Under standard assumptions (local generator, microcausality, linear CPTP, spacelike support separation), operational no-signaling holds: Alice's marginal outcome probabilities are invariant under Bob's spacelike-separated interventions.

VI. TELEOLOGY: MEASURE TILT (COMPANION, OPTIONAL)

All results in Sections 2–5 hold in the $\eta \to 0$ limit; the measure tilt constitutes a companion hypothesis subject to independent falsification.

The canonical route, when retained, is a tilt of the trajectory/outcome measure:

$$\frac{\mathrm{d}P_\eta}{\mathrm{d}P_0}[\Gamma] = \exp\Big(\eta J_E[\Gamma] - \psi(\eta)\,T\Big), \tag{11}$$

with local probability deformation (modified effective decoherence weighting)

$$P_\eta(i) = \frac{p_i\, e^{\eta E_i}}{\sum_j p_j\, e^{\eta E_j}}, \qquad E_i = \frac{\Delta E_i}{C_E}. \tag{12}$$

This is implemented within the CPTP/GKSL container as a measure-level tilt; trace preservation and no-signaling are preserved. Standard quantum mechanics is recovered when $\eta \to 0$, $\Delta E_i \to 0$, or $C_E \to \infty$.

VII. H2 INTERFEROMETRIC VISIBILITY — FLAGSHIP OBSERVABLE

The highest-payoff next physical signal is interferometric visibility suppression. MQGT-SCF predicts a bounded deviation $\delta V \equiv V_0 - V$ from standard decoherence models: the observable is

$$\frac{V}{V_0} = \exp\Big(-\Gamma\, T\, \Delta x^2\Big), \tag{13}$$

where V is observed visibility, V_0 the baseline (no excess channel), T the superposition time, and Δx the path separation. Standard quantum mechanics predicts $\Gamma = 0$; any statistically significant excess above the noise floor defines the test signal.

A null result at total nuisance budget δ_{total} sets the exclusion floor

$$\Gamma_{\text{floor}} = \frac{|\ln(1-\delta_{\text{total}})|}{T\,\Delta x^2}. \tag{14}$$

Null detection at sensitivity δ_{total} therefore excludes any parameter region for which the predicted coupling strength yields $\Gamma > \Gamma_{\text{floor}}$. With the reference planning profile $T = 10^{-6}$ s, $\Delta x = 10^{-3}$ m, $\delta_{\text{total}} \approx 1.15 \times 10^{-3}$,

$$\Gamma_{\text{floor}} \approx 1.15 \times 10^9\ \mathrm{s}^{-1}\,\mathrm{m}^{-2}.$$

A Phase-0 benchtop optical visibility pilot measures the achieved nuisance-budget floor; a null is scientifically valuable and excludes the corresponding region of parameter space.

VIII. EXPERIMENTAL PRIORITY ORDER

- **H2** — Interferometric visibility suppression: primary physical blade.
- **H1** — Estimator and adversarial stress-test channel: simulation-primary, protocol hardening.
- **H3** — Joint consistency / model comparison: inference layer combining H2 with fifth-force, collider, cosmology.

QRNG is retained only as an optional secondary comparator, not the primary path.

IX. DECOUPLING LIMIT

The decoupling limit is

$$\kappa, \gamma, \alpha_H, \beta_H, g_{\Phi_c}, g_E, \eta, \gamma_{\text{col}} \longrightarrow 0 \quad \Longrightarrow \quad \text{GR+SM + standard QM}$$

All predictions of the core EFT reduce to standard GR+SM in the decoupling limit of the scalar sector.

X. WHAT IS SOLVED AND WHAT REMAINS

Solved (formal specification): Minimal EFT action, bounded vacuum condition, covariant CPTP/GKSL measurement dynamics with micro-causality, no-signaling under stated assumptions, decoupling limit.

Remaining: Operationalization of estimators, numerical constraint-fitting, external replication, and decisive empirical confirmation. The program does not justify claims of consciousness/ethics/teleology effects empirically detected, or that the framework has beaten standard QM/GR in live data.

Interpretive layer (non-essential to formalism): Broader philosophical or semantic readings of Φ_c and E are deferred to companion work. The present formulation restricts to their role as dynamical scalar degrees of freedom within a consistent, falsifiable effective field theory and open-system framework.

Appendix A: Claim Ledger v1

A. Core — keep in main physics papers

1. Minimal scalar-singlet EFT extension of GR+SM. Status: Core/active. Evidence: formal ansatz. Demotion: no admissible stable parameter region, no decoupling limit, or redundancy with trivial field redefinition.
2. Standard unitary microdynamics retained. Status: Core/active. Demotion: if foundational nonunitarity sneaks into bulk dynamics.
3. One lawful local covariant CPTP/GKSL measurement container. Status: Core/active; operator-completion is a live task. Demotion: inability to specify operator families preserving CPTP, locality, no-signaling.
4. H2 interferometric visibility suppression as flagship falsifier. Status: Core/active. Demotion: no lab-facing nuisance budget, no achievable Γ_{floor}, or nulls wiping out viable parameter region.
5. H1 as simulation-primary stress-test channel. Status: Core-supporting. Demotion: if rhetorically upgraded into hardware anomaly claim.
6. H3 as executable cross-channel inference layer. Status: Core-supporting. Demotion: if treated as "nature prefers the theory" before external replication.

B. Companion — box separately

7. Optional measure tilt / ethics-weighted Born deformation. Status: Companion only. Demotion: fails normalization, sign, or no-signaling checks.
8. Semantic interpretation of Φ_c and E as consciousness/ethics sectors. Status: Companion/interpretive. Demotion: if no operational observable distinguishes semantic reading from generic singlet-scalar.
9. Jhāna attractors, QRNG ethical-context effects. Status: Companion/exploratory. Demotion: non-predictive or outperformed by standard models.
10. UV uniqueness / SO(10)/string branch. Status: Companion/exploratory. Demotion: if allowed to bleed into current test-facing claims.

C. Implementation-only — useful, not evidence

11. Zora architecture. Status: Implementation layer. Demotion: if presented as physical evidence.
12. Replication bundle, runbooks, manifests. Status: Implementation. Demotion: if existence of code treated as confirmation.

D. Deprecated — retire from mainline

13. Teleology as explicit time-odd bulk source term in Lagrangian.
14. Foundational modified Born rule as required core doctrine.
15. QRNG as primary flagship blade.
16. "Solved empirically," "decisive external evidence," or "final global posterior" language.

Appendix B: Attack Matrix v1

1. **Core EFT claim.** Support: clean EFT formulation, stable potential, decoupling limit. Attack: show ill-defined, redundant, unstable. Kill: vacuum instability, no decoupling limit.
2. **Decoupling claim.** Support: explicit parameter limits. Attack: demand exact decoupling in equations. Kill: residual observable contradicting experiment.
3. **Collapse modification claim.** Support: fully specified master equation, normalization, no-signaling. Attack: exact Lindblad operators, domains, CPTP proof. Kill: violates no-signaling, positivity, interference bounds.
4. Φ_c **physicality claim.** Support: clear observable or operational definition. Attack: Φ_c is relabeling of complexity/synchrony. Kill: all Φ_c effects reproducible by standard neuroscience.

5. **Ethical scalar claim.** Support: reproducible operationalization. Attack: arbitrariness if E depends on human interpretation. Kill: E cannot be measured independently of theory-laden assumptions.

6. **Empirical QRNG claim.** Support: preregistered trials, independent replication. Attack: statistical fishing, drift, hardware bias. Kill: large blinded preregistered null.

7. **Neurophenomenology claim.** Support: state classification predicting EEG/MEG. Attack: mapping is poetic. Kill: standard neural models fit equally well.

8. **Zora/recursive agent claim.** Support: runnable architecture with ablation studies. Attack: does standard RL/Bayesian do the same? Kill: ablations show no benefit.

9. **Falsifiability claim.** Support: short list of outcomes forcing revision. Attack: every negative absorbed by "coupling too small." Kill: no parameter region preserves novelty and survives constraints.

10. **Publication-grade claim.** Support: clean separation of theorem/ansatz/interpretation. Attack: category errors. Kill: overstates what is shown.

Appendix C: Explicit Failure Conditions

- If blinded QRNG experiments show no deviation above threshold X after N trials, the collapse-bias coupling is ruled out in regime R.
- If no operator set can be given that is completely positive and no-signaling, the collapse extension is rejected.
- If Φ_c/E observables cannot be distinguished from standard neural or stochastic models, the field interpretation is demoted to metaphor.
- If H2 interferometric null at achieved Γ_{floor} excludes the viable parameter region, the framework is falsified.
- If fifth-force and equivalence-principle experiments close the allowed scalar-coupling window to zero, the portal sector is ruled out.
- If joint fit over H2, fifth-force, collider, and cosmology yields no surviving parameter region, the framework is overconstrained.

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